us to major gaps in our theories of particle physics. We don't know yet what the solution will be, but it's a safe bet that our explorations of the cosmos will play a role in sorting it out. Uniting cosmology and particle physics has already allowed us to measure the basic shape of spacetime, take an inventory of the components of reality, and peer back through time to an era before the existence of stars and galaxies in order to trace our origins, not just as living beings, but as matter itself.

Of course, it goes both ways. As much as modern cosmology informs our understanding of the very, very small, particle theories and experiments can give us insight into the workings of the universe on the largest scales. This combination of a top-down and bottom-up approach ties into the essence of physics. As much as pop culture would have you believe that science is all about eureka moments and spectacular conceptual reversals, advances in our understanding come more often from taking existing theories, pushing them to the extremes, and watching where they break. When Newton was rolling balls down hills or watching the planets inch across the sky, he couldn't possibly have guessed that we'd need a theory of gravity that could also cope with the warping of spacetime near the Sun, or the unimaginable gravitational forces inside black holes. He would never have dreamed that we'd someday hope to measure the effect of gravity on a single neutron.[fn4] Fortunately, the universe, being really very big, gives us a lot of extreme environments to observe. Even better, it gives us the ability to study the early universe, a time when the entire cosmos was an extreme environment.

A quick note about terminology. As a general scientific term, *cosmology* refers to the study of the universe as a whole, from beginning to end, including its components, its evolution over time, and the fundamental physics governing it. In *astrophysics*, a cosmologist is anyone who studies really distant things, because (1) that means looking at quite a lot of universe and (2) in astronomy, faraway things are also far in the past, since the light that reaches us from them has been traveling for a long time—sometimes billions of years. Some astrophysicists explicitly study the evolution or early history of the universe, while some specialize in distant objects (galaxies, clusters of galaxies, and so forth) and their properties. In *physics*, cosmology can veer in a direction that is much more theoretical. For instance, some cosmologists in physics departments (as opposed to astronomy departments) study alternative formulations of particle physics